EXP_2026Q2_INVALID_SPLIT · VERDICT REPORT · ENGINEER

GROWTH · PRICING · 2026-06-05 · 10,000 UNITS · 14 DAYS

Invalid assignment.

The assignment ratio drifted past the χ^2 threshold. No metric value was read.

TL;DR

INVALID ASSIGNMENT. **Pause and audit the assignment service before reading any metrics.**

The observed assignment split fell outside the χ^2 threshold. The lift number you'd compute would be confounded with the assignment imbalance, so the tree refuses to report it.

ASSIGNMENT IMBALANCE

4,229 / 5,771

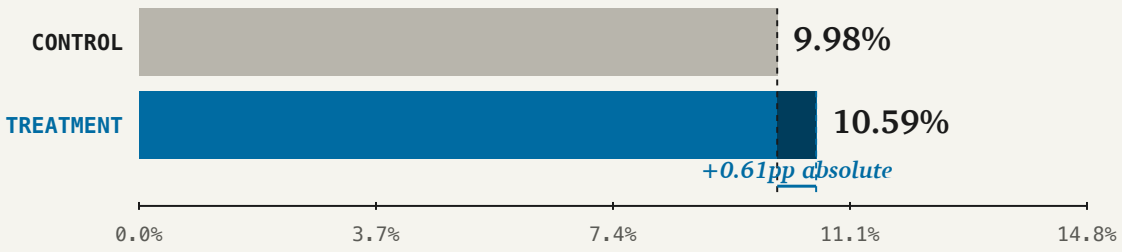
control / treatment — expected near 50 / 50

X^2 STATISTIC	P-VALUE (SRM)
237.78	0.0000
THRESHOLD	**DECISION STEP**
0.0005	step 1

PRIMARY METRIC · CONVERSION_RATE

Treatment 10.59% vs Control 9.98%

Treatment and control rates on the same axis. The dark blue segment on the treatment bar is the absolute lift; the dotted line shows the lift relative to the control end.

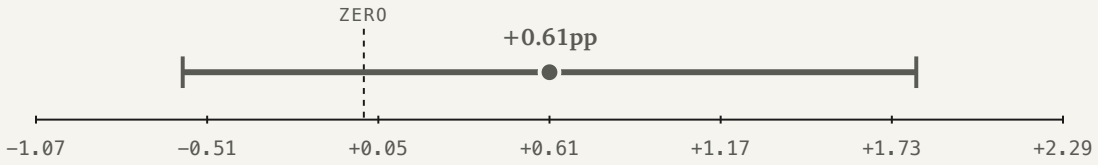


Per-arm conversion. The lift is +0.61pp absolute (+6.1% relative).

CONFIDENCE INTERVAL ON LIFT

95% CI on lift: (-0.59, +1.81)pp — straddles zero.

Zero is inside the interval. We can't reject the null at $a = 0.05$. The verdict tree then asks whether we had enough sample to detect what we cared about.



The CI bar shows the range of plausible true lifts. The crossbars are the 95% bounds; the dot is the point estimate.

GUARDRAILS

Guardrail clear.

The pre-registered guardrails are revenue and (where wired) latency. None crossed their harm thresholds.

revenue_per_user

`threshold ≥ −5.0% · higher_is_better`



`90% CI [−1.17, +8.83]% · realized +3.83% ·` CLEAR

DECISION TREE · HOW THE VERDICT WAS REACHED

Walked 8 steps; step 1 fired the verdict.

Each row is a checkpoint in the verdict tree. Walking stops at the row marked FIRED; that step's logic is what produced the verdict above.

0	Required inputs present (UNVERIFIABLE guard)	`all non-null`
1	SRM passes (or override resolved) · **FIRED**	`srm_pass = False`
2	Guardrails clear (no 90% CI on harm side)	`guard_lift = +3.83%`
3	Powered (n_observed ≥ n_required)	`n = 10,000 / 8,000`
4	Primary CI excludes 0	`CI [−0.59, +1.81] pp`
5	Primary CI not on harm side	`direction = higher_is_better`
6	Lift magnitude ≥ MDE/2	`|lift| = 0.609pp vs MDE/2 = 0.100pp`
7	Novelty stable (late_ratio ≥ 0.7)	`late_ratio = —`

PROVENANCE · AUDITABLE CHAIN

Reproducible from sealed brief + warehouse fixture.

Every quantitative claim above resolves to a row in the warehouse, a function in `agentxp.stats.*`, or a step in `agentxp.interpret.tree`. The chain hash below covers the brief, every stat result, and the verdict tree's terminal step.

EXPERIMENT_ID	exp_2026q2_invalid_split
CHAIN_HASH	34a07f2277a6237d52ae2a16164f8d176eb1ace2283d5d593e705192c71e411e
RENDER_STATUS	VERIFIED
AGENTXP_VERSION	0.3.0-v3
GENERATED_AT	2026-06-05T15:42:04Z
PRIMARY_METRIC	conversion_rate · proportion_test
SRM_TEST	srm_chi_squared · threshold 0.0005
TREE_MODULE	agentxp.interpret.tree.walk_tree
REPRODUCE	python scratch/walk_experiment.py exp_2026q2_invalid_split

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